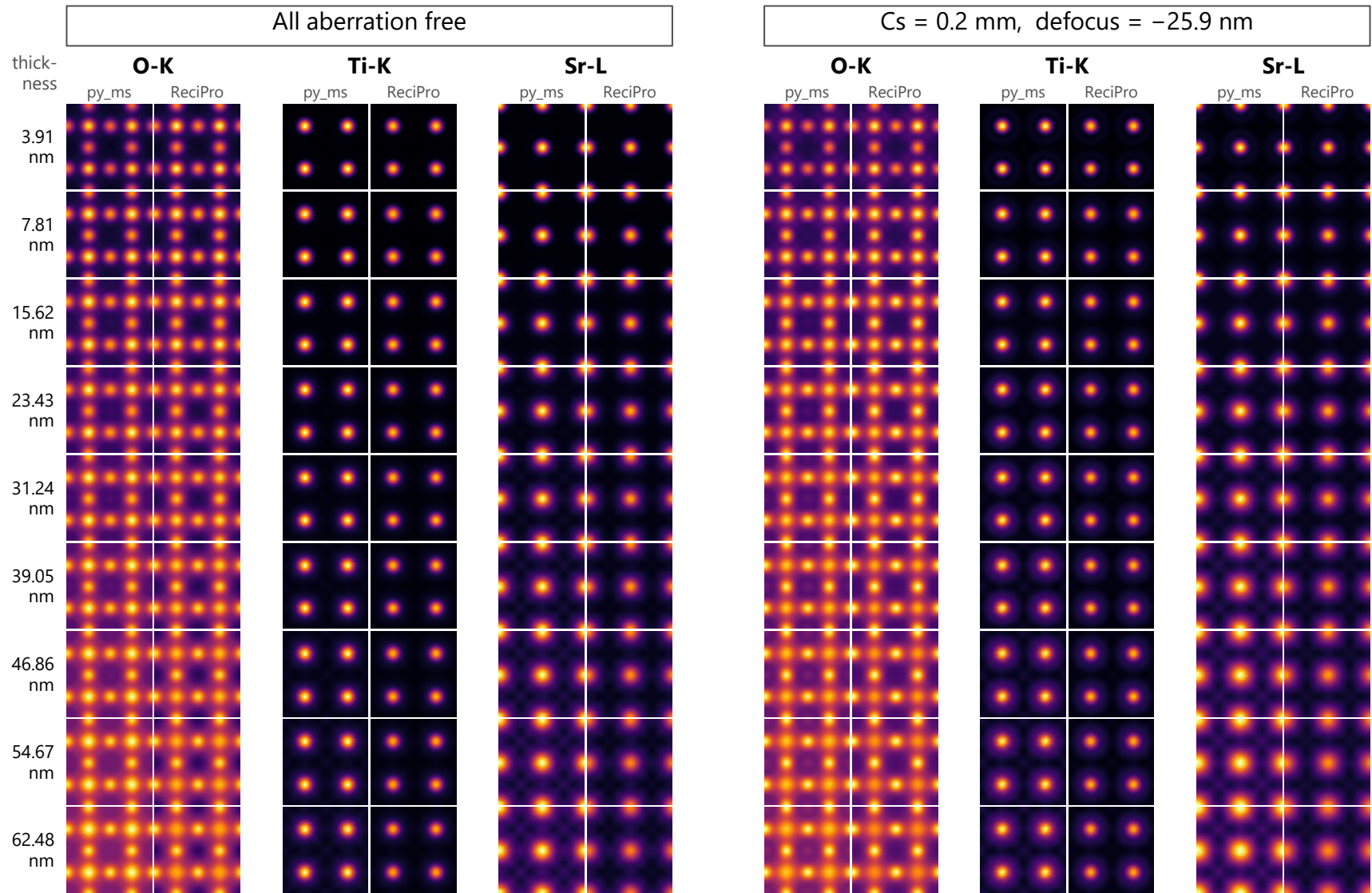


# Comparison of STEM-EDX simulations by py\_multislice (5e5e4d8) and ReciPro (v4.945)

Sample: SrTiO<sub>3</sub> [001] ( $B = 8\pi^2\langle u^2 \rangle$ : Sr 0.621, Ti 0.440, O 0.732 Å<sup>2</sup>)  
Acc. vol.: 200 kV, beam convergence: 15 mrad  
Signal: inner-shell ionizations per incident electron; Sr-L = L1+L2+L3

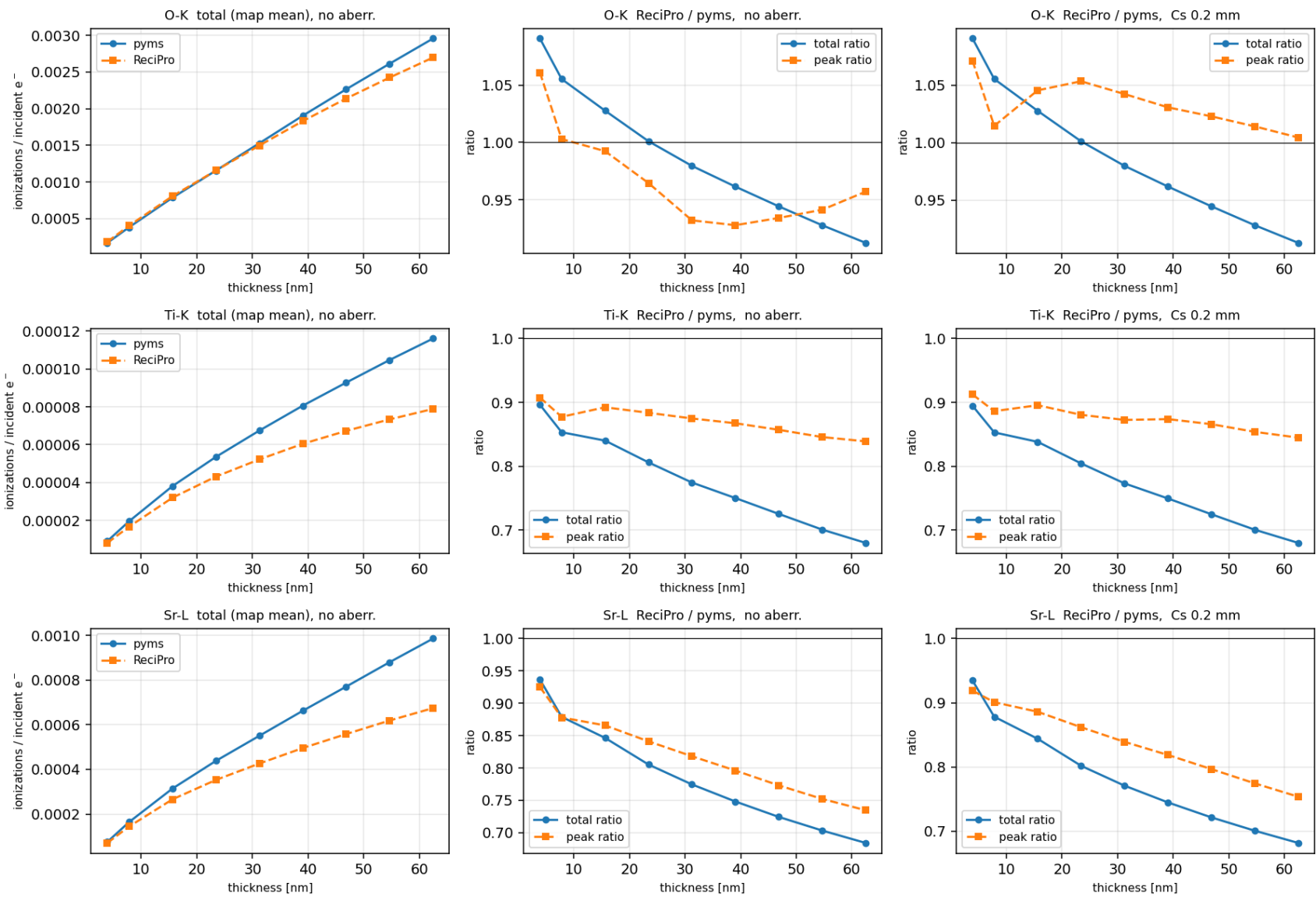
py\_multislice conditions: multislice, frozen phonon (25 configurations), 6×6×1 supercell, 384<sup>2</sup> grid, slice a/2, Allen-group parameterised ionization potentials (Sr = 2p + 2s).  
ReciPro conditions: Bloch wave method, 400 waves, 64<sup>2</sup> probe directions,  $\Delta z = 0.195$  nm, DHFS-KS23 F(s) + Bote-Salvat  $\sigma$  (dataset v1.0.0).



Each panel: one unit cell scanned 32×32 (shown 2×2 tiled). Colour scale is shared within each py\_multislice/ReciPro pair, so absolute differences are visible.

# Quantitative comparison

STEM-EDX quantitative comparison: ReciPro vs py\_multislice (SrTiO<sub>3</sub> [001], 200 kV, 15 mrad)

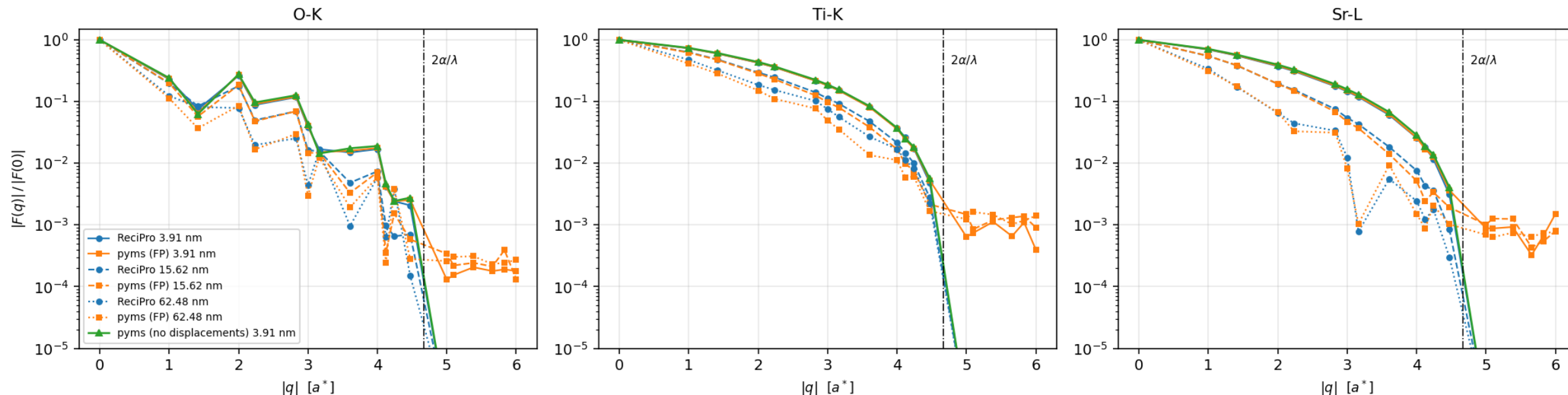


|             | $\sigma$ ratio<br>pyms / ReciPro<br>( $t \rightarrow 0$ ) | shape agreement<br>$ F(q) $ ratio, 3.9 nm<br>( $q \leq 3 \text{ \AA}^{-1}$ ) |
|-------------|-----------------------------------------------------------|------------------------------------------------------------------------------|
| <b>O-K</b>  | 0.92                                                      | 0.96 – 1.11                                                                  |
| <b>Ti-K</b> | 1.12                                                      | 0.995 – 1.007                                                                |
| <b>Sr-L</b> | 1.07                                                      | 0.978 – 0.990                                                                |

- Normalized map shapes agree to 1–2 % (Ti-K, Sr-L) in the thin limit; the same holds with Cs = 0.2 mm.
- Totals differ by the cross-section source: Bote–Salvat (ReciPro) vs the Allen-group tables (pyms),  $\pm 10\text{--}17\%$  depending on the shell.
- The ReciPro/pyms total falls with thickness (Ti-K 0.90  $\rightarrow$  0.68 at 62 nm): ReciPro’s absorptive model removes thermally scattered electrons, while frozen phonon keeps them ionizing. This quantifies the practical error of the absorptive approximation for EDX.
- Baseline against ionization dataset v1.0.0; the Sr-L entries are expected to move with the Dirac-recomputed L tables (v2).

# Spatial-frequency content — band limit and frozen-phonon aliasing

Spatial-frequency content of the EDX maps (aberration free). ReciPro is exactly zero beyond  $2\alpha/\lambda$  as it must be; py\_multislice's out-of-band content is frozen-phonon aliasing.



- Both codes are strictly band-limited: the EDX map has no spatial frequency beyond  $2\alpha/\lambda = 4.67 a^*$  at 15 mrad, as required for incoherent STEM imaging — ReciPro and py\_multislice (thermal displacements off) both drop to zero there.
- The residual out-of-band floor ( $\sim 10^{-3}$ ) of the frozen-phonon curves is aliasing of the  $6 \times 6$  supercell phonon configurations; it disappears when thermal displacements are disabled and is invisible at map level.
- Within the band the normalized Fourier amplitudes of the two codes track each other over three orders of magnitude.